



## Research Paper

## Physiology-guided artificial intelligence digital twins for simulating glucose dynamics in type 2 diabetes

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## ABSTRACT

**Background:** Metabolic simulators help accelerate the development of diabetes technologies, but existing mechanistic models for type 2 diabetes (T2D) are difficult to individualize and provide limited explanation of the complex interplay between physiology and available treatments. **Methods:** First, we trained NeuralOGTT, a neural network state-space model that replicates glucose–insulin dynamics during the oral glucose tolerance test (OGTT) in T2D. Next, we determined the minimal model architecture using a variance-based sensitivity analysis. Finally, we created digital twins (DT-NeuralOGTT) by adding auxiliary sub-networks to the NeuralOGTT trained using individuals data from a free-living observational study (N = 38, age: 50 ± 10 years; 27 females; 1 high risk of diabetes, 9 pre-T2D, and 28 T2D). We simulated study scenarios and assessed the digital twins ability to replicate real-world glucose outcomes: time in range (TIR, 70 – 180 mg/dL), time in tight range (TITR, 70 – 140 mg/dL), time below range (TBR, < 70 mg/dL), and time above range (TAR, > 180 mg/dL). **Results:** NeuralOGTT replicated glucose dynamics during OGTTs. Personalized DT-NeuralOGTT enabled generalization under free-living conditions across 12-h simulations, achieving simulated glucose outcomes that are statistically identical to observed outcomes: TIR: 91.8 (95% CI: 82.2–96.0)% vs. 92.1 (95% CI: 83.5–95.8)%, TITR: 56.0 (95% CI: 36.6–77.4)% vs. 59.3 (95% CI: 48.7–76.4)%, TAR: 6.3 (95% CI: 1.6–15.8)% vs. 7.4 (95% CI: 3.8–15.7)%, and TBR: 0.5 (95% CI: 0.0–1.0)% vs. 0.1 (95% CI: 0.1–0.2)%. **Conclusion:** NeuralOGTT models glucose–insulin dynamics using a structure that preserves explainability while requiring small amounts of training data. DT-NeuralOGTT incorporates individual variability to better capture dynamics in pre-T2D and T2D. Code available: [https://github.com/mosquera-lopez/T2DSim\\_AI](https://github.com/mosquera-lopez/T2DSim_AI)

## 1. Introduction

Type 2 diabetes (T2D) is a chronic condition that affects glucose metabolism through either an increase in insulin resistance or a reduction in pancreatic insulin production, both resulting in the need for glycemic self-management to maintain a healthy, euglycemic range of 70–180 mg/dL [1,2]. Achieving euglycemia is burdensome: requiring continuous monitoring of glucose levels, maintaining a healthy diet, regular physical activity [3–5], and in some cases, administering glucose-lowering medications [6,7]. Suboptimal T2D management can lead to serious microvascular and macrovascular complications [1,8].

In order to reduce the burden of T2D management, there is a growing interest in adapting clinical decision support systems [9–11] and automated drug delivery systems [12], which have helped to reduce the burden of glucose management in type 1 diabetes (T1D), to the management of T2D [13]. Central to developing these technologies – both models inside of control algorithms, and processes for *in-silico* evaluation – are glucose–insulin dynamics simulators which replicate the body's glucose response to meals, physical activity, medications, and stress.

Broadly, these simulators fall into three key classes. First, we have mechanistic models such as those utilizing ordinary differential equations (ODEs). These are the most common approaches [14–20], due to

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**Table 1**  
Summary of existing modeling approaches for type 2 diabetes (T2D).

| Study                    | Aim                                                                                                                                                                               | Modeling Approach                                                                                   | Limitations                                                                              |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Zhang et al., [22]       | Forecast disease progression (e.g., HbA1C, eGFR, glucose, etc) using multiomic data and clinical features                                                                         | Machine learning models and knowledge graphs                                                        | Limited longitudinal data and sample size; primarily tested on healthy participants      |
| Shamanna et al., [23]    | Provide guidelines that would enable individual patients to avoid foods that cause blood glucose spikes and to replace them with foods that do not produce spikes.                | Daily precision nutrition guidance based on CGM, food intake data, and machine learning algorithms. | Tested in a small population; digital twin construction methods unclear.                 |
| Thamotharan et al., [24] | Personalize diabetes treatment for elderly T2D patients, optimizing insulin infusion.                                                                                             | Deep learning models, mechanistic models and model predictive control.                              | Tested only in insulin-dependent population.                                             |
| Jauslin et al., [15]     | Model the glucose–insulin system using oral glucose tolerance test (OGTT) data                                                                                                    | Mechanistic ODE model.                                                                              | Applied only to OGTT data; not validated for real-world or free-living glucose dynamics. |
| Silber et al., [19]      | Model the glucose–insulin system using intravenous data.                                                                                                                          | Mechanistic ODE model.                                                                              | Validated only on intravenous glucose provocations.                                      |
| Visentin et al., [25]    | Model the glucose–insulin system using triple-tracer mixed meal test data.                                                                                                        | Mechanistic ODE model.                                                                              | Limited real-world testing.                                                              |
| Vahidi et al., [18]      | Model the glucose–insulin system using OGTT and isoglycemic intravenous glucose infusion test data.                                                                               | Mechanistic ODE model.                                                                              | Limited real-world testing.                                                              |
| Ahdab et al., [20]       | A model for glucose–insulin dynamics, which incorporates the effect of multiple glucose meal sizes, different metformin doses, physical activity, insulin injections, and stress. | Mechanistic ODE model.                                                                              | Not tested with real world data.                                                         |

their ability to provide interpretable glucose dynamics. However, these models often fall short in capturing individual physiological phenomena and rarely fit real-world data accurately. Second, data-driven and artificial intelligence (AI) models enable improved personalization, but require a large amount of data which can be difficult to obtain, and lack interpretability, even to the point of misrepresenting the glucose dynamics [21]. Between these two classes lie physiology-guided AI models, which integrate physiological knowledge with data-driven methods. This emerging class of models offers a promising alternative by combining the explainability of ODE-based models with the personalization capabilities of AI, helping overcome the limitations of each individual approach.

Our work lies in this third class: the proposal of a novel neural network (NN) physiology-guided AI digital twin framework that enables flexible, explainable, and personalized metabolic modeling of the glucose–insulin system. To better contextualize existing approaches, we provide a structured summary of representative prior work, including their aim, modeling approach, and limitations, in Table 1.

The primary objective of this study is to develop a physiology-guided AI framework capable of learning accurate and interpretable glucose–insulin dynamics in individuals with pre-T2D and T2D from limited data, while enabling personalization under free-living conditions. We hypothesize that combining physiological structure with data-driven learning improves the ability to simulate real-world glucose outcomes compared to purely mechanistic or purely data-driven approaches.

Our key contributions are as follows:

- 1. Novel OGTT Model for pre-T2D and T2D (NeuralOGTT):** A NN state-space model trained using ODE-based aggregate supervision to approximate general glucose–insulin dynamics during the oral glucose tolerance test (OGTT) in a population with pre-T2D and T2D. NeuralOGTT can simulate the correct glucose–insulin response (i.e., a rise in glucose and insulin following a glucose load, followed by a decline), even with limited training data, providing a foundation for further personalizing the simulator to individual-level data.
- 2. Novel AI Model Training Techniques:** A variance-based sensitivity analysis to determine the minimal architecture required to achieve accuracy and explainable glucose–insulin dynamics, along with a

novel loss function balancing personalization and alignment with known physiology.

- 3. Personalized Digital Twins for T2D (DT-NeuralOGTT):** A virtual patient population of digital twins that can replicate the glucose outcomes of individuals with pre-T2D and T2D under free-living conditions. These digital twins leverage the baseline NeuralOGTT to capture general glucose–insulin dynamics and individual-level sub-networks to improve simulation accuracy by integrating additional glucose management and contextual data (e.g., heart rate, sleep data, medications, and time-related features). This framework enables modeling of inter- and intra-individual variability in glucose dynamics, while reducing the requirement for significant amounts of data from any single individual.

## 2. Materials and methods

### 2.1. Dataset

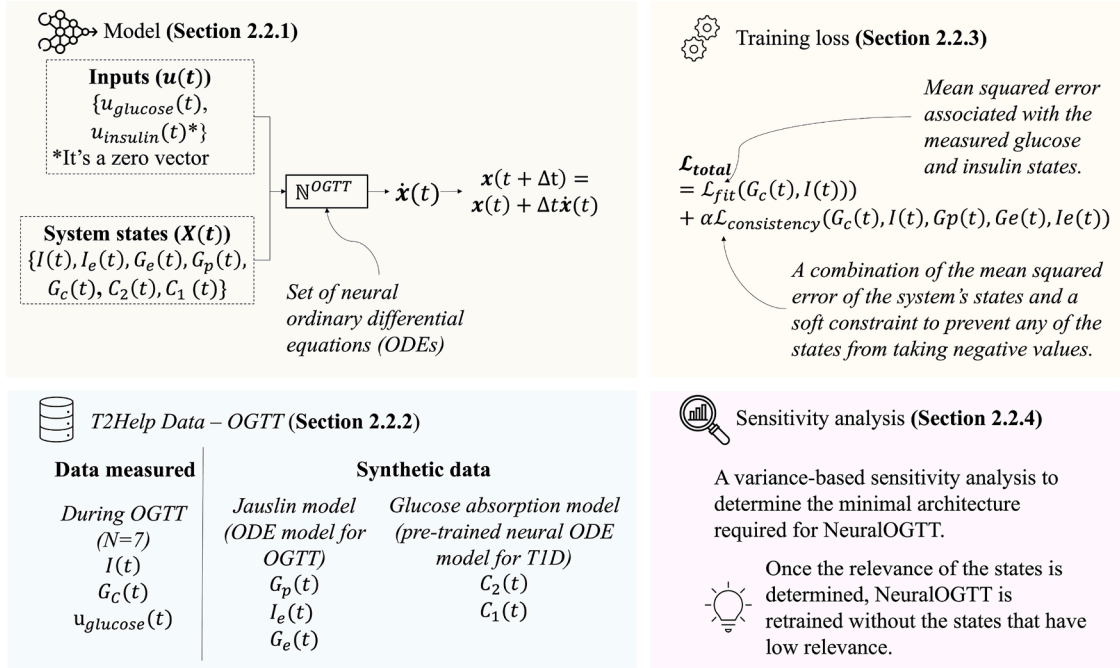
Models were developed utilizing data from the T2Help dataset, along with synthetic data generated by multiple simulation models. T2Help is a multi-site observational study that collected free-living data during 16 weeks from adults at risk of diabetes, pre-T2D, and T2D (NCT04503239). Study participants wore a continuous glucose monitor (CGM) (Dexcom G6 [26]), an activity tracker (Fitbit [27]), and filled out an e-diary (WellDoc [28]) documenting their medication and food intake. All participants consented to take part in the study, and the secondary analysis presented herein has been approved by OHSU's Institutional Review Board (IRB 25550).

Due to restrictions in data access, we utilized only the subset of data collected at Oregon Health & Science University. After filtering for data completeness, our dataset included 38 participants with the following characteristics: mean age:  $50 \pm 10$  years; 27 females; mean BMI:  $35 \pm 9$  kg/m<sup>2</sup>; 1 at high risk of diabetes, 9 with pre-T2D, and 28 with T2D; 27 White, 7 Asian, and four other races/ethnicities. We refer to this dataset as *T2Help Data – Free Living*.

In addition to the CGM, Fitbit, and WellDoc data, we were able to obtain OGTT data for training the NeuralOGTT model for a subset of seven participants who had a positive C-peptide laboratory result (mean

## NeuralOGTT (Section 2.2)

Glucose-insulin model during oral glucose tolerance test (OGTT) in a population with pre-T2D and T2D



**Fig. 1.** Overview of the NeuralOGTT framework. This schematic summarizes key components of the NeuralOGTT model presented in Section 2.2, which simulates glucose-insulin dynamics during an oral glucose tolerance test (OGTT) in individuals with pre-T2D and T2D.

age  $45 \pm 7$  years; 5 females; mean BMI  $33 \pm 5 \text{ kg/m}^2$ ; 1 pre-T2D, and 6 T2D; 4 White, 2 Asian, and 1 other race/ethnicity). We refer to this subset as *T2Help Data – OGTT*. These participants were instructed to fast overnight and not take any medication before the OGTT.

To bolster the real patient data from *T2Help Data – OGTT*, a variety of well-established glucose-insulin simulation models in the literature were utilized, including the Jauslin model [15], and the meal absorption compartments from a neural ODE simulator [29] based on the Hovorka model [30].

### 2.2. NeuralOGTT: glucose-insulin dynamics during OGTT

NeuralOGTT is a neural network ODE-based model designed to simulate typical glucose-insulin dynamics during an OGTT in individuals with pre-T2D and T2D. NeuralOGTT is not intended to learn population-level physiological mechanisms, nor to replace mechanistic descriptions of glucose metabolism. Instead, it aims to learn a physiologically consistent baseline representation of the OGTT response that can subsequently be adapted to individual-level data. By embedding established physiological dynamics within a neural state-space formulation, NeuralOGTT provides a flexible foundation for integration with individualized digital-twin models while maintaining consistency with prior physiological knowledge. A model overview is presented in Fig. 1, with details herein.

#### 2.2.1. Model

We utilize a hybrid modeling approach that integrate NNs with state-space modeling to construct a model comprising seven sub-networks (known as neural ODEs [31]). Each sub-network approximates a first-order ODE that represents a state in the glucose dynamics model, including (1) the Jauslin model [15], for T2D glucose-insulin kinetics and dynamics, and (2) the carbohydrate absorption states from Hovorka et al. [30] (see Supplementary Material S1). The ODEs ensure adherence to known physiological dynamics, while the NN component allows per-

sonalization and the potential to capture metabolic factors not explicitly included in the ODE models.

The compact structure of NeuralOGTT is presented in Eq. (1), and Eqs. (2a)–(2g) show the explicit definitions for each component. For ease of readability, we will use the compact notation throughout the paper otherwise noted.

$$\dot{\mathbf{x}}(t) = \mathbf{N}^{\text{OGTT}}(\mathbf{x}(t), \mathbf{u}(t); \Theta^{\text{OGTT}}) \quad (1)$$

$$\dot{C}_1(t) = \mathbb{N}_{C_1}(C_1(t), u_{\text{glucose}}(t); \theta_{C_1}) \quad (2a)$$

$$\dot{C}_2(t) = \mathbb{N}_{C_2}(C_1(t), C_2(t); \theta_{C_2}) \quad (2b)$$

$$\dot{G}_c(t) = \mathbb{N}_{G_c}(G_c(t), G_p(t), C_2(t), I_e(t); \theta_{G_c}) \quad (2c)$$

$$\dot{G}_p(t) = \mathbb{N}_{G_p}(G_c(t), G_p(t); \theta_{G_p}) \quad (2d)$$

$$\dot{G}_e(t) = \mathbb{N}_{G_e}(G_c(t), G_e(t); \theta_{G_e}) \quad (2e)$$

$$\dot{I}_e(t) = \mathbb{N}_{I_e}(I(t), I_e(t); \theta_{I_e}) \quad (2f)$$

$$\dot{I}(t) = \mathbb{N}_I(I(t), C_2(t), G_2(t), u_{\text{insulin}}(t); \theta_I) \quad (2g)$$

In Eq. (1),  $\mathbf{x}(t) = \{x^i(t)\} = \{C_1(t), C_2(t), G_c(t), G_p(t), G_e(t), I_e(t), I(t)\}$  are the system's states at time  $t$ ;  $\mathbf{u}(t) = \{u_{\text{glucose}}(t), u_{\text{insulin}}(t)\}$  the system's inputs; and  $\mathbf{N}^{\text{OGTT}} = \mathbb{N}_{x^i}$  is a set of fully connected NNs parameterized by  $\Theta^{\text{OGTT}} = \theta_{x^i}$ , with  $i = 1, 2, \dots, 7$ . Note that  $G_c$  and  $I$  are measured states, representing the central glucose and insulin concentrations in plasma, respectively; the remaining states are non-measured. Definitions of all states and model inputs are provided in Table 2.

#### 2.2.2. T2Help data – OGTT preparation

Plasma glucose and insulin concentrations were measured every 15 min over a 315-min period (23 samples per participant). A 75-g glucose load was administered at minute 15. Data were linearly interpolated to obtain points at 5-min intervals.

For the supervised training of NeuralOGTT, we used a combination of measured data from the OGTT (i.e.,  $G_c$  and  $I$ ), which are measured states, and synthetic data for unmeasured states (i.e.,  $G_e$ ,  $I_e$ , and  $G_p$ )

**Table 2**  
Description of the system's states and inputs of NeuralOGTT.

| Name                       | Description                                 | Units    |
|----------------------------|---------------------------------------------|----------|
| <b>System's States</b>     |                                             |          |
| $x^1 = C_1$<br>$x^2 = C_2$ | Glucose absorption from the gut into plasma | Unitless |
| $x^3 = G_c$                | Central glucose (Measured)                  | mg/dL    |
| $x^4 = G_p$                | Peripheral glucose                          | mg       |
| $x^5 = G_e$                | Effect of glucose on insulin secretion      | mg/dL    |
| $x^6 = I_e$                | Effect of insulin on glucose disposal       | mU/L     |
| $x^7 = I$                  | Insulin in plasma (Measured)                | mU/L     |
| <b>System's Inputs</b>     |                                             |          |
| $u_{glucose}$              | Carbohydrate intake                         | g        |
| $u_{insulin}$              | Insulin on board                            | U/h      |

generated using the model described by Jauslin et al. [15] driven by the observed glucose and insulin measurements (see Supplementary Material S2 for a detailed description of the synthetic data generation process). Importantly, these latent state estimates were generated independently for each participant using their own observed OGTT measurements. Consequently, the resulting training dataset reflects the variability present in the observed OGTT data rather than a single synthetic OGTT trajectory, as illustrated in Supplementary Fig. 2. For the glucose absorption states (i.e.,  $C_1$  and  $C_2$ ), we generated the estimates of the states during the OGTT using a pre-trained neural ODE model [29].

While the only input is a glucose load at  $u_{glucose}$  ( $t = 15 \text{ min} = 75 \text{ g}$ ), we chose to include  $u_{insulin}$  as a zero input in the model to enable ease-of-use in contexts where exogenous insulin is delivered.

Each NeuralOGTT system's state is scaled using a robust scaler to reduce the effect of outliers on the model training (see Supplementary Material S3). The final development dataset  $\mathcal{D}^{OGTT} = \{\mathbf{x}(t), \mathbf{u}(t)\}$  consists of estimates of the system's states and inputs to the model during the OGTT. Data was split into training ( $\mathcal{D}_{train}^{OGTT}$ ) and validation ( $\mathcal{D}_{validation}^{OGTT}$ ) subsets using the leave-one-participant-out cross-validation technique.

### 2.2.3. NeuralOGTT training process

We utilize the truncated simulation error minimization methodology described by Forgione and Piga [32] to train the NeuralOGTT. Given  $\mathcal{D}_{train}^{OGTT}$ , we create batches containing  $B$  sequences of  $M$  data points. We find the optimal set of parameters  $\Theta^{OGTT}$  that minimized the loss function defined in Eq. (3) that is used to embed physiological knowledge into NeuralOGTT.

$$\mathcal{L}_{total} = \mathcal{L}_{fit} + \alpha \mathcal{L}_{consistency} \quad (3)$$

Where  $\mathcal{L}_{fit}$  is the loss associated with the measured glucose and insulin states  $G_c$  and  $I$  (Eq. (4)),  $\mathcal{L}_{consistency}$  is the loss associated with all trainable system's states (Eq. (5)), and  $\alpha > 0$  is a regularization parameter empirically selected to balance these two terms. The training objective prioritizes agreement with measured data through  $\mathcal{L}_{fit}$ , while  $\mathcal{L}_{consistency}$  acts as a soft constraint encouraging physiologically plausible dynamics in non-observed states.

$$\mathcal{L}_{fit} = \sum_{i=3,7} \left[ \frac{1}{BM} \sum_{b=1}^B \sum_{m=1}^M (x_{b,m}^{i,sim} - x_{b,m}^i)^2 \right], \quad (4)$$

$$\mathcal{L}_{consistency} = \frac{1}{5} \sum_{i=3}^7 \left[ \frac{1}{BM} \sum_{b=1}^B \sum_{m=1}^M (x_{b,m}^{i,sim} - x_{b,m}^i)^2 + \beta \sum_{b=1}^B \sum_{m=1}^M \max[0, -(x_{b,m}^{i,sim} - \min(x^i))] \right] \quad (5)$$

In Eqs. (4) and (5),  $x_{b,m}^{i,sim}$  reflects the simulated  $i$ th state value in the timestamp  $m$  for the batch  $b$  and  $x_{b,m}^i$  reflects the actual respective state value. In Eq. (5), the first component of the consistency loss is the mean squared error (MSE) between  $x_{b,m}^{i,sim}$  and  $x_{b,m}^i$ . The second component is a soft constraint regularized by the parameter  $\beta \geq 0$  added to prevent the states from taking values that are less than those observed in the training set. This type of constraint helps the model learn to avoid simulating unrealistic values.

Supported by evidence that carbohydrate absorption dynamics are broadly comparable between T2D and T1D populations [17], we leverage models that were previously pre-trained in individuals with T1D [29]. In particular, the sub-networks corresponding to  $C_1$  and  $C_2$  were pre-trained on data from individuals with T1D and fine-tuned at the end of NeuralOGTT's training. This enables us to leverage a larger dataset while accounting for differences in carbohydrate absorption between T1D and T2D. Notably, the pre-trained model is based on the Hovorka carbohydrate absorption formulation [30], thereby preserving its physiological structure while enabling seamless integration into our neural ODE framework and efficient fine-tuning without retraining these compartments from scratch. Consequently, in Eq. (5), the summation runs from  $i = 3$  to  $i = 7$ , as  $C_1$  ( $i = 1$ ) and  $C_2$  ( $i = 2$ ) are excluded from the loss function.

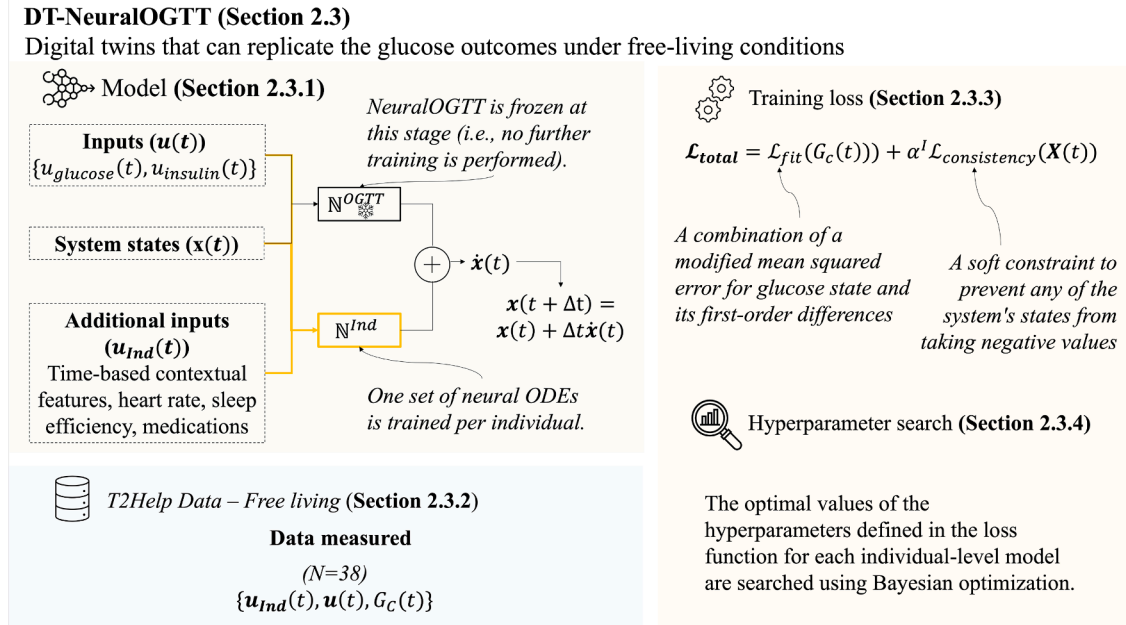
We selected a simple architecture for each sub-network  $\mathbb{N}_{x_i}$  which corresponds to a NN of one hidden layer with a rectified linear unit (ReLU) activation function. The model was trained with the Adam optimizer [33], using the following empirically determined hyper-parameters: batch size  $B = 2$ , sequence length (corresponding to the duration of the OGTT)  $M = 64$  with sampling period  $\Delta t = 5 \text{ min}$ , regularization constants  $\alpha = 1$  and  $\beta = 1.3e^{-3}$ , the number of neurons for each trainable sub-network is 64, starting learning rate  $\lambda = 10^{-4}$  scheduled to be reduced to  $10^{-5}$  after 300 epochs, and a total of 350 epochs. Due to the limited availability of OGTT sequences (i.e., data from 7 participants), we utilize a small batch size of  $B = 2$ .

### 2.2.4. Sensitivity analysis-driven NeuralOGTT minimal architecture

We define the minimal NeuralOGTT as the NN state-space model with the smallest set of states that can reproduce the same output as the proposed model. This approach offers significant advantages, including requiring fewer trainable parameters while preserving the interpretability.

To determine the minimal NeuralOGTT, we aim to identify compartments that do not significantly influence the model output. Specifically, the first-order derivative of plasma glucose,  $\dot{x}^3$ , during the OGTT. For this purpose, we apply the Sobol method, a variance-based global





**Fig. 2.** Overview of the DT-NeuralOGTT framework. This figure illustrates the key components of DT-NeuralOGTT (2.3), a physiology-guided digital twin model designed to replicate glucose dynamics under free-living conditions.

sensitivity analysis technique, which decomposes the variance of a model output and attributes it to a given set of inputs.

We adapt the method described by Escorihuela-Altaba, et.al. [34] to estimate the Total Sobol Index ( $TSI$ ), which quantifies the contribution of each state  $x^i$ ,  $i = 1, \dots, 7$ , to the overall variance of the model output,  $\dot{x}^3$ . To assess the influence of each state  $x^i$ , we introduced a weighting factor  $\omega_{x^i}$  that directly scales the neural ODE associated with that state. This was done by modifying the output of each neural ODE to  $\dot{x}^i := \omega_{x^i} \dot{x}^i$ , with  $\omega_{x^i} \in [0.1, 1.9]$  corresponding to a perturbation range of  $-90\%$  to  $+90\%$  of the derivative results. We utilize these limits as expanding the range did not lead to significant changes in the results. The  $TSI$  was then calculated for each  $\omega_{x^i}$  at each simulation time, quantifying the proportion of the total variance in  $\dot{x}^3$  attributed to perturbations in the  $\dot{x}^i$  compartment.

To calculate  $TSI$ , we used the SALib Python library [35,36] following the methodology described by Norton [37].  $TSI$  for a  $\omega_{x^i}$  at time  $t$  is defined in Eq. (6).

$$TSI_{\omega_{x^i}}(t) = \frac{\text{Var}_t[\dot{x}^3 | \omega_{x^i}]}{\text{Var}_t[\dot{x}^3]}, \quad (6)$$

In Eq. (6),  $\text{Var}_t[\dot{x}^3 | \omega_{x^i}]$  represents the variance of the model output,  $\dot{x}^3$ , at time  $t$ , conditional on the weighting factor  $\omega_{x^i}$  and  $\text{Var}_t[\dot{x}^3]$  is the total variance of  $\dot{x}^3$  at time  $t$ .  $TSI_{\omega_{x^i}}(t)$  ranges from 0 to 1: a value of 1 indicates that the variance in the model output is entirely explained by  $\omega_{x^i}$  and its interactions with other compartments, whereas a value of 0 indicates no influence on the output variance.

Therefore, the relative importance of each state  $x^i$  is inferred from the  $TSI_{\omega_{x^i}}(t)$  values, as the associated weighting factor directly modulates its dynamics within the neural ODE. A high  $TSI_{\omega_{x^i}}(t)$  indicates that perturbations to the state  $x^i$  have a significant impact on the variance of  $\dot{x}^3$ , making it essential for accurate simulation of plasma glucose dynamics. Conversely, low  $TSI_{\omega_{x^i}}(t)$  values suggest that the corresponding state contributes minimally to the output variance and could be considered for simplification or removal from the model without significant impact on overall dynamics.

A qualitative analysis of the variance-based sensitivity results for each state was performed to determine which states could be removed from NeuralOGTT. Once the new minimal model architecture was im-

plemented, NeuralOGTT was trained, removing the states that have little influence on plasma glucose.

### 2.3. DT-NeuralOGTT: Virtual patient population for personalized glucose dynamics modeling

DT-NeuralOGTT is a digital twin framework for simulating glucose dynamics under free-living conditions. This framework combines the typical glucose–insulin dynamics learned by NeuralOGTT with individualized compartments trained using free-living data and incorporating other signals not accounted for in conventional physiologic models. The learning process of the NeuralOGTT parameters was guided by a well-established physiological model with fixed parameters such that NeuralOGTT captures a baseline physiologically consistent glucose–insulin response to a carbohydrate load. Inter- and intra-individual variability are introduced at this stage through personalization using real-world data. An overview of the model structure, training process, and used dataset is shown in Fig. 2.

#### 2.3.1. Personalized models: From NeuralOGTT to digital twins

While the NeuralOGTT model describes general glucose–insulin dynamics in response to an OGGT glucose load, it fails to account for inter- and intra-individual variability during free-living conditions. To address this, we augmented the NeuralOGTT model with a set of trainable individual-level neural ODEs  $N^{\text{Ind}}$  (Eq. (7)) parametrized by individual-specific parameters  $\Theta^k$ ,  $k = 1, \dots, K$ , where  $K$  is the number of individuals. This results in digital twin models (DT-NeuralOGTT) that can be personalized via gradient descent with minimal data and updated as data availability increases.

$$\dot{x}'(t) = N^{\text{Ind}}(x(t), u(t), u_{\text{Ind}}(t); \Theta^k) \quad (7)$$

$N^{\text{Ind}}$  has the same inputs and states as  $N^{\text{OGTT}}$ . The individual model also has additional inputs ( $u_{\text{Ind}}(t)$ ) which include sleep efficiency, heart rate, medication information, and contextual time information (e.g., hour of day, weekday vs. weekend) specific to a given individual and which are not represented in the ODE models. To capture individual patterns, we empirically designed each sub-network of  $N^{\text{Ind}}$  as a single-layer NN with 256 neurons and a ReLU activation function. In this way,

the simulation of a state from  $t$  to  $t + \Delta t$  of a digital twin  $k$  is defined in Eq. (8).

$$\mathbf{x}(t + \Delta t) = \mathbf{x}(t) + (\mathbf{N}^{\text{OGTT}}(\mathbf{x}(t), \mathbf{u}(t); \Theta_{\text{OGTT}}) + \mathbf{N}^{\text{Ind}}(\mathbf{x}(t), \mathbf{u}(t), \mathbf{u}_{\text{Ind}}(t); \Theta^k))\Delta t \quad (8)$$

The explicit definition of  $\mathbf{N}^{\text{Ind}}$  will be presented in the results following the definition of the sensitivity analysis-driven architecture reduction.

### 2.3.2. T2Help data – free living preparation

The dataset used to train each DT-NeuralOGTT, denoted as  $D_k^{\text{Ind}} = \{\mathbf{x}(t), \mathbf{u}(t), \mathbf{u}_{\text{Ind}}(t)\}$ , was generated using the *T2Help Data – Free Living* dataset. Apart from the actual glucose measured by the CGM ( $x^3(t) = G_c(t)$ ), all system states ( $\mathbf{x}(t)$ ) were simulated using Eq. (8) during the training of the digital twin. The input  $\mathbf{u}(t)$  was obtained from self-reported meals and exogenous insulin events, which were included only for individuals who were insulin-dependent; otherwise, a zero-input vector was used for insulin. The individual input vector ( $\mathbf{u}_{\text{Ind}}(t)$ ) included the following inputs:

1. **Timed-based contextual features:** Hour of day coded as  $\cos(2\pi \text{ hour}/24)$ ,  $\sin(2\pi \text{ hour}/24)$ ; and a binary variable denoting weekend vs. weekday.
2. **Heart rate:** Heart rate measurements expressed in beats per minute (BPM) recorded with a Fitbit fitness watch. To better characterize short-term physiological dynamics, within each 5-min window we compute summary statistics including the mean, minimum, maximum, and standard deviation of heart rate.
3. **Sleep efficiency:** Fraction of time within a 5-min window spent sleeping as recorded using a Fitbit watch.
4. **Medications:** Medication intake is initially recorded as binary events for each class (SGLT-2, GLP-1, biguanides, and sulfonylurea). These impulse signals are subsequently transformed into continuous inputs using a simple mechanistic kinetic model (see Supplementary Material S4), allowing for a more realistic representation of medication effects over time.

Heart rate and medications was scaled using a robust scaler as done for the states in the NeuralOGTT. The other inputs are inherently between -1 and 1 and do not require scaling. In cases where the participant was insulin-dependent, instead of specifying the insulin amount at a specific time, the input was the insulin on board over 8 h, calculated using the insulin kinetics subsystem from the Hovorka model [30]. This approach was necessary because our model does not include the delay compartments required to represent exogenous insulin kinetics.

The dataset,  $D_k^{\text{Ind}}$ , was split in a 60:20:20 ratio for training ( $D_{\text{train},k}^{\text{Ind}}$ ), validation ( $D_{\text{validation},k}^{\text{Ind}}$ ) and testing ( $D_{\text{test},k}^{\text{Ind}}$ ), respectively. Details on data exclusion and other considerations can be found in Supplementary Material S5.

### 2.3.3. DT-NeuralOGTT training process

The models constructed for all digital twins were trained following the methodology outlined in the NeuralOGTT training process section above, with the following minor updates described herein.

Given  $D_{\text{train},k}^{\text{Ind}}$ , we created batches of size  $B^I$  containing 12-h sequences ( $M^I = 144$  data points) with 98% overlap between consecutive sequences. The overlap was used for training data augmentation. Because we penalized each timestamp in the sequence, we required sequences to have 100% of the CGM values present to be included in the training phase. For the training phase, we only retained 12-h sequences that included at least one reported meal event, as this input is the most likely to affect glucose variability. The loss function (Eq. (9)) for individual models is analogous to that of NeuralOGTT in that it balances

the loss associated with the measured glucose  $G_c$  ( $\mathcal{L}_{\text{fit}}^I$ ) and the loss associated with all system's states ( $\mathcal{L}_{\text{consistency}}^I$ ):

$$\mathcal{L}_{\text{total}}^I = \mathcal{L}_{\text{fit}}^I + \alpha^I \mathcal{L}_{\text{consistency}}^I \quad (9)$$

As in NeuralOGTT,  $\alpha^I \geq 0$  is a regularization parameter selected to balance  $\mathcal{L}_{\text{fit}}^I$  and  $\mathcal{L}_{\text{consistency}}^I$ .

$\mathcal{L}_{\text{fit}}^I$  is a combination between MSE of the simulated 12-h sequences and the actual glucose data, and the MSE first-order differences of the simulated 12-h sequences and the actual glucose data (Eq. (10)). These two terms are balanced by a regularization parameter  $\beta^I \geq 0$ . The first term is multiplied by a penalty function,  $\mathcal{P}(x^{3,\text{sim}}, x^3)$  (Eq. (11)), which was designed to penalize the errors made by the model on infrequent yet clinically significant events, such as hypoglycemia (glucose  $< 70$  mg/dL) and extreme hyperglycemia (glucose  $> 250$  mg/dL); both of which pose challenges in glucose management for individuals with T2D [38,39]. The optimal values for the penalty function  $p_{\text{hypo}}$  and  $p_{\text{hyper}}$  were optimized for each individual using a Bayesian search. This procedure will be described in Section 2.3.4.

$$\mathcal{L}_{\text{fit}}^I = \frac{1}{B^I M^I} \sum_{b=1}^{B^I} \sum_{m=1}^{M^I} \mathcal{P}(x_{b,m}^{3,\text{sim}}, x_{b,m}^3) (x_{b,m}^{3,\text{sim}} - x_{b,m}^3)^2 + \frac{\beta^I}{B^I M^I} \sum_{b=1}^{B^I} \sum_{m=2}^{M^I} ((x_{b,m}^{3,\text{sim}} - x_{b,m-1}^{3,\text{sim}}) - (x_{b,m}^3 - x_{b,m-1}^3))^2 \quad (10)$$

$$\mathcal{P}(x^{3,\text{sim}}, x^3) = \begin{cases} p_{\text{hypo}} & x^{3,\text{sim}} \leq 70 \wedge x^3 > y \\ p_{\text{hyper}} & x^{3,\text{sim}} \geq 250 \wedge x^3 < y \\ 1 & \text{otherwise} \end{cases} \quad (11)$$

$\mathcal{L}_{\text{consistency}}^I$  (Eq. (12)) is defined as a soft constraint to prevent any of the system's states from taking negative values.  $n_{\text{states}}$  is defined as the number of states in the system.

$$\mathcal{L}_{\text{consistency}}^I = \frac{1}{n_{\text{states}}} \sum_{i=1}^{n_{\text{states}}} \sum_{b=1}^{B^I} \sum_{m=1}^{M^I} \max[0, -x_{b,m}^{i,\text{sim}}] \quad (12)$$

For training each individual-level model, we used a learning rate of  $\lambda^I = 10^{-5}$ , regularization constants  $\alpha^I = 10^{-4}$ , and  $\beta^I = 10^2$ , a batch size  $B^I = 32$ , and each individual model training was trained for 25 epochs.

To ensure no bias between DT-NeuralOGTT and the base NeuralOGTT, we perform the following: first, for the individuals with OGTT data, we ensure none of their OGTT data was used in the NeuralOGTT model on which their DT-NeuralOGTT is based; second, during the training process of each DT-NeuralOGTT, the NeuralOGTT model components are frozen (i.e., no updating of weights).

### 2.3.4. Hyperparameter search

Bayesian optimization [40] is used to search for the optimal penalty values of  $p_{\text{hypo}}$  and  $p_{\text{hyper}}$  for each individual-level model, from a range of [1,100]. The optimization objective was to improve similarity between simulated and actual glucose, as measured by root mean squared error (RMSE) and percentage time in clinically significant glucose ranges on the validation set  $D_{\text{validation},k}^{\text{Ind}}$ . Each individual model for the hyperparameter search was trained for 25 epochs. A unique optimization was performed for each individual.

## 2.4. Simulation of real-world scenarios using DT-NeuralOGTT digital twins

### 2.4.1. Estimating initial states

To initialize the simulator when only the initial CGM data were available, we estimated the set of initial states based on steady-state assumptions for all states except  $G_c$ . For  $G_c$ , we used the CGM reading at time 0 along with the trend over the preceding 30 min, estimated using a linear fit. Using these parameters, we searched for the roots of the NeuralOGTT model output. During each iteration of this root-finding process, we monitored whether the states remained within known physiological ranges and capped them when necessary to avoid starting with

non-physiologic values. With this procedure, we obtain a reliable estimate for  $\mathbf{x}(0)$ .

#### 2.4.2. Conducting simulations

Given the initial states  $\mathbf{x}(0)$ , inputs  $\mathbf{u}(t)$  and  $\mathbf{u}_{\text{Ind}}(t)$ , and  $t = 0, \dots, T$ , a  $T$ -length simulation of the glucose dynamics of the  $k^{\text{th}}$  digital twin parameterized by  $\Theta_k$  can be performed using Euler's ODE integration method shown in Eq. (8). The algorithm for this procedure is provided in Supplementary Material S6.

#### 2.5. Evaluation of simulation accuracy

##### 2.5.1. Evaluation of NeuralOGTT

To evaluate NeuralOGTT's ability to replicate an OGTT, we first performed a variance-based sensitivity analysis as described in Section 2.2.4 and retrained the model after removing compartments that did not meaningfully contribute to plasma glucose variability. Given the small number of OGTT sequences of the *T2Help Data – OGTT* dataset, we then visually inspected the glucose and insulin dynamics throughout the OGTT test. Finally, we assessed the similarity between the full NeuralOGTT model (with seven compartments) and the reduced model by comparing their RMSE values against the measured glucose levels during the OGTT. To avoid biasing the results, the glucose value at time zero was excluded from the error calculation, as it was used to initiate the simulation.

##### 2.5.2. Evaluation of DT-NeuralOGTT

**Digital twins approaches used as comparator models:** To benchmark the proposed digital twin framework, we compared its performance against two alternative modeling approaches: a data-driven model and a mechanistic model. Both approaches were individualized per participant and calibrated using the same training and validation datasets, while evaluation was conducted on the testing set.

- **Data-driven model:** We implemented an individual-specific Support Vector Regression (SVR) model to predict the glucose increment:

$$\Delta G(t) = G(t + \Delta t) - G(t). \quad (13)$$

The SVR model takes as input a history of glucose measurements along with contextual features:

$$\Delta G(t) = \text{SVR}(G(t - 6\Delta t), G(t - 5\Delta t), G(t - 4\Delta t), G(t - 3\Delta t), G(t - 2\Delta t), G(t - \Delta t), G(t), \mathbf{u}_{\text{Ind}}(t)), \quad (14)$$

where 30 min of past glucose data ( $\Delta t = 5$  min) were provided to ensure consistency with the initialization procedure used in our proposed framework.

To simulate glucose trajectories on the testing set, predictions were generated recursively according to:

$$G(t + \Delta t) = G(t) + \Delta G(t), \quad (15)$$

where each predicted value  $G(t + \Delta t)$  is fed back as input for subsequent predictions.

The individual input vector  $\mathbf{u}_i^{\text{ind}}$  included insulin on board (for individuals receiving exogenous insulin); carbohydrate availability to approximate the compartment-based absorption models; heart rate features (mean and variability); and sleep-related features. Time and medication variables were not included as inputs because preliminary experiments showed that their inclusion led the model to learn the mean glucose behavior rather than capture the underlying dynamics.

The SVR was trained using a radial basis function (RBF) kernel, as it empirically yielded more stable solutions that better captured glucose dynamics, rather than converging to average behavior. The hyperparameters  $C$ ,  $\gamma$ , and  $\epsilon$ , which respectively control the tradeoff between model complexity and training error, the influence range of individual data points in the RBF kernel, and the tolerance to small prediction error, were optimized via Bayesian optimization.

- **Mechanistic model:** We employed the mechanistic model proposed by Ahdab et al., [20], which integrates multiple physiological subsystems, including glucose-insulin dynamics, carbohydrate intake, metformin pharmacodynamics, and physical activity. This model enables the incorporation of inputs such as carbohydrate intake, heart rate, metformin administration, and exogenous insulin.

Although the full model consists of 120 parameters, prior sensitivity analysis of its underlying formulation [18] identified a subset of influential parameters. Based on this analysis, we optimized 18 parameters per individual: the 15 most sensitive parameters identified in [18], along with 3 additional parameters related to metformin sensitivity. The inclusion of these additional parameters was motivated by empirical observations that population-level values produced unrealistically strong responses to metformin intake.

Individual parameter sets were identified using Bayesian optimization by simulating 12-h glucose trajectories and minimizing prediction error, consistent with the evaluation procedure used for our proposed digital twins.

**Statistical analysis of similarity between simulated and real-world glucose outcomes:** While RMSE is appropriate for assessing pointwise agreement during the NeuralOGTT development, where the objective is to replicate the glucose trajectory from a structured task, it is less informative for evaluating free-living CGM data. In this context, small phase shifts or noise differences can inflate RMSE without reflecting meaningful discrepancies in physiological behavior. For this reason, we evaluated similarity using clinically interpretable CGM-derived metrics. To assess how well the digital twins replicated clinically relevant glucose outcomes, we compared average glucose metrics between simulated and real-world profiles. Specifically, we evaluated time in range (TIR, 70 – 180 mg/dL), time in tight range (TITR, 70 – 140 mg/dL), time below range (TBR, < 70 mg/dL), and time above range (TAR, > 180 mg/dL), as well as high/low blood glucose indices (HBGI and LBGI) and mean glucose (MG). These metrics are widely used to assess clinical performance and glucose control [41], all standard measures of glucose control [41].

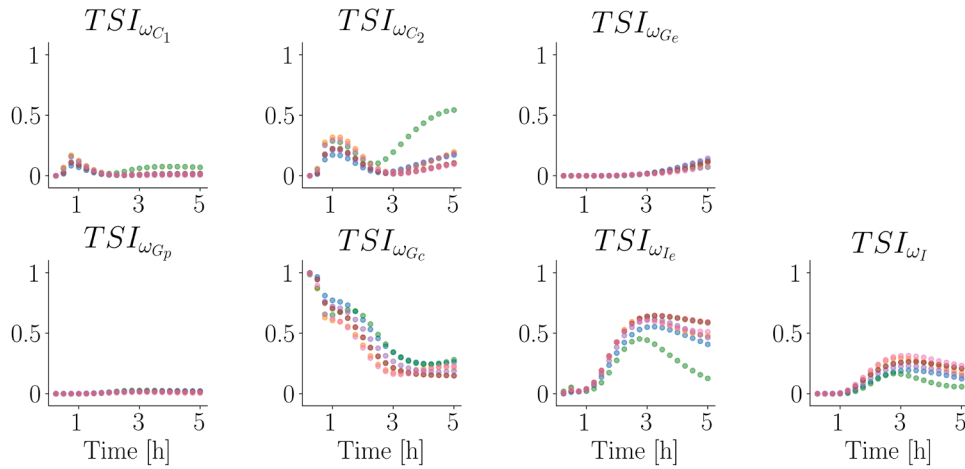
Because the analysis is based on a relatively small sample size ( $N = 38$ ) and most metrics were non-normally distributed (see Supplementary material S7), we used a non-parametric, equivalence-based approach to assess similarity between the simulated population and the *T2Help Data – Free Living*. Specifically, subject-level differences in glucose outcomes between simulated and observed data were computed and summarized using bootstrapping of the median differences. The resulting confidence intervals (CIs) were evaluated against predefined equivalence margins, and replication was considered successful when these CIs lay within the margins in both directions. This approach avoids reliance on p-values, which may be unreliable in small, non-normally distributed samples.

We also estimated the 95% CIs of the median glucose outcomes for both real and simulated data. The 95% CIs were obtained using a bootstrapping procedure with replacement, drawing samples of size 50 across 10,000 iterations. We expect the 95% CIs of the real and simulated medians to be comparable. The equivalence margins for each metric were:  $\delta_{\text{TIR}} = 5\%$ ,  $\delta_{\text{TITR}} = 5\%$ ,  $\delta_{\text{TAR}} = 5\%$ ,  $\delta_{\text{TBR}} = 1\%$ ,  $\delta_{\text{HBGI}} = 1$ ,  $\delta_{\text{LBGI}} = 0.5$  and  $\delta_{\text{MG}} = 10$  mg/dL [29,42,43].

These metrics were reported on the testing set ( $\mathcal{D}_{\text{test},k}^{\text{Ind}}$ ) using non-overlapping 12-h sequences, and compared across the three approaches: the data-driven model, the mechanistic model, and the proposed DT-NeuralOGTT framework. Time zero was excluded from the calculation of all metrics, as it was used to initialize the simulation.

**Parkes error grid:** To evaluate the clinical accuracy of the virtual patient population as an *in-silico* simulation tool, we used the Parkes error grid [44] to compare the point-based simulated and real-world glucose levels over 12-h sequences from the testing set.

**Additional analysis of simulation error:** To further demonstrate the potential of digital twins as a reliable *in-silico* simulation tool, we



**Fig. 3.** Variance-based sensitivity analysis in terms of Total Sobol Index (TSI) for each state, which helped guide minimal model design. Each color represents an individual oral glucose tolerance test (OGTT) from the *T2Help Data - OGTT*.

computed the distribution of the mean RMSE over different segment lengths: from 30 min up to 12-h. For each time segment included in the RMSE calculation, we allow a maximum of 30% CGM sensor drop out.

**Contribution of individual-level inputs:** To assess the contribution of each component of the individual input vector  $\mathbf{u}_{\text{Ind}}(t)$ , we trained multiple versions of the digital twin model using different subsets of the input features. Specifically, we trained models using only one input at a time - heart rate-based features, medications, time-based features-as well as a model without additional inputs (i.e.,  $\mathbf{u}_{\text{Ind}}(t) = \{\}$ ). They were compared with the proposed NeuralOGTT digital twin.

Each model configuration was evaluated across three key physiological events known to impact glucose levels (i.e., meals, physical activity, and sleep). We used the same outcome metrics described above (TIR, TITR, TAR, TBR, HBGI, LBGI, and MG). This experiment aimed to determine whether incorporating all individual-level inputs provides added value and improves simulation accuracy compared to models trained with partial or no individual-level information.

**Evaluation of digital twins behavior under variable inputs:** We evaluate the ability of the DT-NeuralOGTT digital twins to reproduce glucose dynamics under varying input conditions by simulating controlled perturbations of key inputs. Specifically, we select representative scenarios from the testing set and systematically modify individual inputs while keeping all others fixed.

For meal-related scenarios, we identify instances with isolated meal events and simulate standardized carbohydrate intakes of 30 g, 60 g, and 90 g, ensuring no additional meals occur within one hour before and two hours after the event.

For physical activity, we consider events grouped by duration (10–40 min, 40–90 min, and >90 min). In these cases, the original scenarios are simulated, and additional simulations are performed by replacing heart rate-based features with baseline values to assess the impact of physical activity in the glucose response.

For medication-related scenarios, we select events where a participant takes a given medication and perform simulations both with and without the corresponding input signal over a 12-h window. In the case of exogenous insulin, we specifically analyze bolus insulin events associated with meals.

All simulations are visualized to compare the resulting glucose and insulin responses across the digital twins. These analyses qualitatively demonstrate the ability of the digital twins to capture the effects of meal size, physical activity, and medication on glucose dynamics.

### 3. Results

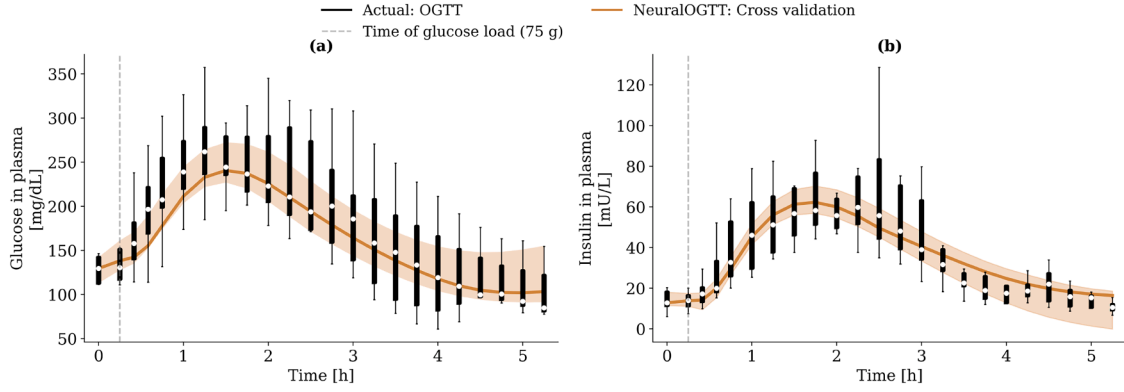
#### 3.1. NeuralOGTT: variance-based sensitivity analysis and minimal model

We utilized  $TSI$ -based variance sensitivity analysis to guide the NeuralOGTT minimal model design. Sensitivity analysis for each state in the NeuralOGTT model (Eq. (2)) is presented in Fig. 3. We find varying  $\dot{C}_2(t)$ ,  $\dot{I}_e(t)$ , and  $\dot{I}(t)$  within the  $\pm 90\%$  range increases the variability of  $\dot{G}_c(t)$  after glucose ingestion, indicating their strong influence on  $\dot{G}_c(t)$ . In contrast, we find  $\dot{G}_p(t)$  has a negligible effect, supporting the exclusion of  $G_p$  from NeuralOGTT. Time-dependent variance can also be observed. Because the simulation begins at steady state, other state differentials do not immediately affect  $\dot{G}_c(t)$ , yielding a high initial  $TSI$  for  $\dot{G}_c(t)$ . As time progresses, glucose ingestion triggers insulin production and we see this reflected in the  $TSI$  of  $\dot{C}_2(t)$ ,  $\dot{I}_e(t)$ , and  $\dot{I}(t)$  rising while that of  $\dot{G}_c(t)$  declines. Fig. 3 also shows lower relevance for  $\dot{C}_1(t)$ ,  $\dot{G}_e(t)$ , and  $\dot{G}_p(t)$ . We find although  $\dot{C}_1(t)$  may still matter as a delay state,  $\dot{G}_p(t)$  does not contribute meaningfully to the  $\dot{G}_c(t)$  variability, and  $\dot{G}_e(t)$  may only become important under other disturbances such as exercise or stress.

The difference in RMSEs between the minimal NeuralOGTT model (6 states) and the originally proposed model (7 states) was negligible during the OGTT scenario ( $45.2 \pm 10.5$  mg/dL vs.  $45.3 \pm 8.7$  mg/dL). Couple with the sensitivity analysis, this indicates strong validity of the minimal model. We note for the general NeuralOGTT, we do not expect a low RMSE. Instead, the goal of this model is to learn the appropriate glucose–insulin dynamics (i.e., a rise in glucose and insulin following a glucose load, followed by a decline), to enable low-data personalization in the DT-NeuralOGTT. Fig. 4 shows the results of simulating the OGTT response (i.e., glucose and insulin in plasma) of the *T2Help Data - OGTT* participants using NeuralOGTT without  $G_p$ . Additional details on the relative contributions of the training loss components are provided in Supplementary Material S8. Results obtained using different values of the consistency-loss weighting parameter  $\alpha$  show that  $L_{\text{fit}}$  is responsible for reproducing the measured glucose and insulin trajectories, whereas  $L_{\text{consistency}}$  promotes physiologically meaningful latent-state dynamics. Together, these terms balance agreement with observed data and physiological interpretability.

Thus, the minimal NeuralOGTT architecture ( $\mathbf{N}^{\text{OGTT}}$ ), derived from the variance-based sensitivity analysis, is shown in Table 3 (column 1), and the individualized models ( $\mathbf{N}^{\text{Ind}}$ ) are shown in column 2.  $\mathbf{N}^{\text{Ind}}$  includes additional inputs whose connections follow their underlying physiological effects, for example, sulfonylureas stimulate pancreatic insulin release [45] and are therefore linked to the Insulin compartment.





**Fig. 4.** Leave-One-Out-Cross-Validation results of the minimal NeuralOGTT implementation (e.g. without  $G_p$ ). Minimal NeuralOGTT results are shown in orange, with the median (thick line) and interquartile range (shaded area). Actual oral glucose tolerance test (OGTT) measurements are shown in black boxplots; the white dots indicate the medians. **(a)** Glucose in plasma compartment. **(b)** Insulin in plasma compartment.

**Table 3**

Sensitivity analysis-driven reduced architecture of the NeuralOGTT model ( $\mathbb{N}^{OGTT}$ ) and the individual-level model ( $\mathbb{N}^{Ind}$ ).

| $\mathbb{N}^{OGTT}(\mathbf{x}(t), \mathbf{u}(t); \Theta_{OGTT})$              | $\mathbb{N}^{Ind}(\mathbf{x}(t), \mathbf{u}(t), \mathbf{u}_{Ind}(t); \Theta^k)$                                                                                           |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\dot{C}_1(t) = \mathbb{N}_{C_1}(C_1(t), u_{glucose}(t); \theta_{C_1})$       | $\dot{C}_1'(t) = \mathbb{N}_{C_1}'(C_1(t), u_{glucose}(t), u_{glp1}(t); \theta_{C_1}^k)$                                                                                  |
| $\dot{C}_2(t) = \mathbb{N}_{C_2}(C_1(t), C_2(t); \theta_{C_2})$               | $\dot{C}_2'(t) = \mathbb{N}_{C_2}'(C_1(t), C_2(t); \theta_{C_2}^k)$                                                                                                       |
| $\dot{G}_e(t) = \mathbb{N}_{G_e}(G_e(t), C_2(t), I_e(t); \theta_{G_e})$       | $\dot{G}_e'(t) = \mathbb{N}_{G_e}'(G_e(t), G_p(t), C_2(t), I_e(t), u_{hr}(t), u_{sleep}(t), u_{weekend}(t), u_{hour}(t), u_{biguanide}(t), u_{sgr12}(t); \theta_{G_e}^k)$ |
| $\dot{G}_e(t) = \mathbb{N}_{G_e}(G_e(t), G_e(t); \theta_{G_e})$               | $\dot{G}_e'(t) = \mathbb{N}_{G_e}'(G_e(t), G_e(t); \theta_{G_e}^k)$                                                                                                       |
| $\dot{I}_e(t) = \mathbb{N}_{I_e}(I_e(t), I_e(t); \theta_{I_e})$               | $\dot{I}_e'(t) = \mathbb{N}_{I_e}'(I_e(t), I_e(t); \theta_{I_e}^k)$                                                                                                       |
| $\dot{I}(t) = \mathbb{N}_I(I_e(t), C_2(t), G_2(t), u_{insulin}(t); \theta_I)$ | $\dot{I}'(t) = \mathbb{N}_I'(I_e(t), C_2(t), G_2(t), u_{insulin}(t), u_{glp1}(t), u_{sulfo}(t); \theta_I^k)$                                                              |

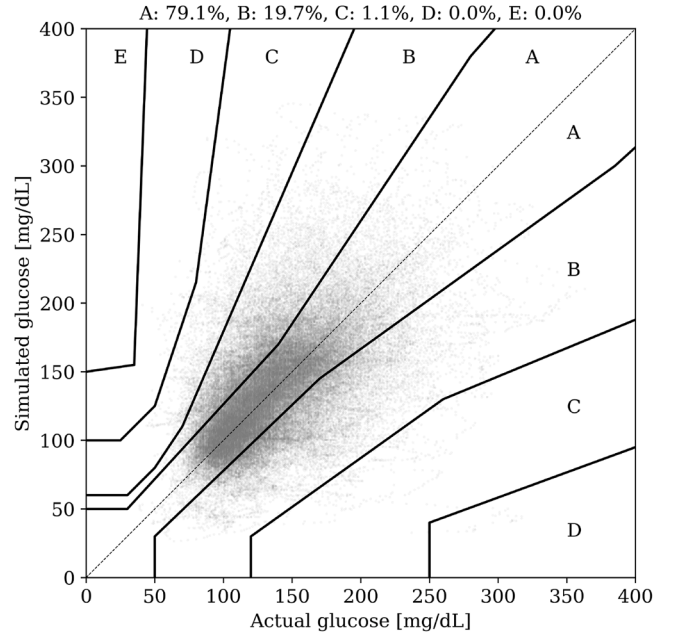
This results in an optimized architecture that reduces the number of trainable parameters while preserving interpretability for NeuralOGTT and DT-NeuralOGTT.

### 3.2. DT-NeuralOGTT: virtual patient population

Utilizing the NeuralOGTT model as a base for OGTT dynamics, we extend the model to simulate a wide range of real-world glucose dynamics through construction of the DT-NeuralOGTT digital twins. This enables the creation of a virtual population of 38 physiology-guided AI digital twins. Table 4 summarizes the statistical comparison between DT-NeuralOGTT, the mechanistic model, and the data-driven model on the testing set. Both DT-NeuralOGTT and the data-driven model successfully met the equivalence criteria for all evaluated glucose metrics, with the exception of T1TR, as indicated by the CIs of both the median values and the paired differences. In contrast, the mechanistic model met the equivalence criteria only for TAR and HBGI. Notably, the mechanistic model exhibited a higher TBR, suggesting an overestimation of glucose-lowering effects, potentially due to excessive sensitivity to exogenous insulin, metformin, or endogenous insulin production.

For 12-h sequence simulations, DT-NeuralOGTT achieved the lowest RMSE ( $31.6 \pm 11.2$  mg/dL), outperforming both the mechanistic model ( $55.6 \pm 16.7$  mg/dL) and the data-driven model ( $34.0 \pm 11.6$  mg/dL).

The Parkes error grid analysis showed that 98.8% of the simulated data for the 38 digital twins fell within zones A and B of the Parkes error grid (Fig. 5), demonstrating high clinical accuracy of the 12-h simulations. Fig. 6 presents the distribution of mean RMSE values across different simulation window durations. The model simulations were excellent for shorter durations up to 3 h in the future with a mean RMSE of less than 20 mg/dL in the 30-min simulations and under 10 mg/dL for simulations out to two hours. Notably, the RMSE of the DT-NeuralOGTT digi-



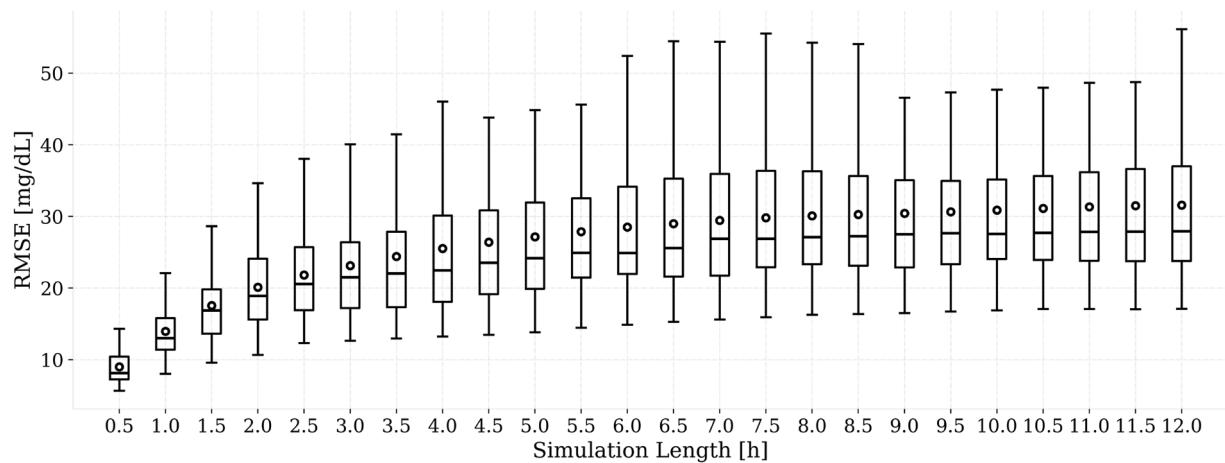
**Fig. 5.** Parkes error grid for 12-h sequences across the 38 digital twins.

tal twins plateaued as the simulation duration increased, indicating that the model maintains consistent accuracy over extended periods. These results are particularly relevant as only a single initialization point is used at  $t = 0$ .

**Table 4**

Bootstrap medians and 95% confidence intervals of glucose outcomes on the test set. For each metric, the top sub-row reports the median (95% CI) of the actual real-world data and the predictions from three models (DT-NeuralOGTT, mechanistic model, and data-driven model), evaluated on 12-h sequences across 37 digital twins. The bottom sub-row reports the mean paired difference (actual - predicted) with its 95% bootstrap CI. Equivalence was assessed using predefined clinically meaningful margins for each metric (TITR:  $\pm 5\%$ , TIR:  $\pm 5\%$ , TAR:  $\pm 5\%$ , TBR:  $\pm 1\%$ , LBGi:  $\pm 0.5$ , HBGI:  $\pm 1$ , mean glucose:  $\pm 10$  mg/dL); bold values indicate glucose outcomes that are equivalent to the actual measurements. Testing was conducted on 37 digital twins for some outcomes because one individual did not meet the 12-h sequence requirements.

|                      |                               | Actual                   | DT-NeuralOGTT<br>(Our approach)       | Mechanistic<br>model             | Data-driven<br>model                  |
|----------------------|-------------------------------|--------------------------|---------------------------------------|----------------------------------|---------------------------------------|
| <b>TIR</b><br>[%]    | Median value<br>[95% CI]      | 92.1<br>[83.5 – 95.8]    | <b>91.8</b><br><b>[82.2, 96.0]</b>    | 63.8<br>[60.9, 67.1]             | <b>89.1</b><br><b>[83.1, 94.3]</b>    |
|                      | Median difference<br>[95% CI] |                          | <b>0.2</b><br><b>[-1.3, 2.6]</b>      | 20.7<br>[18.3, 24.7]             | <b>0.2</b><br><b>[-1.1, 2.6]</b>      |
| <b>TITR</b><br>[%]   | Median value<br>[95% CI]      | 59.3<br>[48.7 – 76.4]    | 56.0<br>[36.6, 77.4]                  | 50.5<br>[44.7, 55.4]             | 56.0<br>[42.2, 70.6]                  |
|                      | Median difference<br>[95% CI] |                          | 4.5<br>[3.2, 8.5]                     | 8.6<br>[2.7, 18.0]               | 4.3<br>[2.0, 9.3]                     |
| <b>TAR</b><br>[%]    | Median value<br>[95% CI]      | 7.4<br>[3.8 – 15.7]      | <b>6.3</b><br><b>[1.6, 15.8]</b>      | <b>6.9</b><br><b>[4.9, 10.0]</b> | <b>9.5</b><br><b>[4.4, 16.1]</b>      |
|                      | Median difference<br>[95% CI] |                          | <b>0.5</b><br><b>[-0.2, 2.1]</b>      | <b>0.0</b><br><b>[-0.3, 1.8]</b> | <b>-0.1</b><br><b>[-1.6, 1.1]</b>     |
| <b>TBR</b><br>[%]    | Median value<br>[95% CI]      | 0.1<br>[0.1 – 0.2]       | <b>0.5</b><br><b>[0.0, 1.0]</b>       | 23.5<br>[20.5, 28.8]             | <b>0.0</b><br><b>[0.0, 0.2]</b>       |
|                      | Median difference<br>[95% CI] |                          | <b>-0.2</b><br><b>[-0.7, 0.0]</b>     | -23.5<br>[-27.3, -20.1]          | <b>0.0</b><br><b>[0.0, 0.1]</b>       |
| <b>LBGI</b>          | Median value<br>[95% CI]      | 0.2<br>[0.1 – 0.3]       | <b>0.4</b><br><b>[0.1, 0.6]</b>       | 6.7<br>[5.1, 9.7]                | <b>0.2</b><br><b>[0.0, 0.3]</b>       |
|                      | Median difference<br>[95% CI] |                          | <b>-0.0</b><br><b>[-0.2, 0.0]</b>     | -6.7<br>[-8.4, -4.6]             | <b>0.0</b><br><b>[-0.0, 0.1]</b>      |
| <b>HBGI</b>          | Median value<br>[95% CI]      | 2.6<br>[1.4 – 3.6]       | <b>2.5</b><br><b>[1.1, 3.9]</b>       | <b>1.5</b><br><b>[1.1, 2.3]</b>  | <b>2.8</b><br><b>[1.6, 3.8]</b>       |
|                      | Median difference<br>[95% CI] |                          | <b>-0.0</b><br><b>[-0.3, 0.2]</b>     | <b>0.3</b><br><b>[0.1, 0.7]</b>  | <b>-0.1</b><br><b>[-0.4, 0.0]</b>     |
| <b>MG</b><br>[mg/dL] | Median value<br>[95% CI]      | 136.4<br>[126.0 – 147.2] | <b>135.1</b><br><b>[124.0, 148.2]</b> | 105.8<br>[99.7, 112.1]           | <b>139.8</b><br><b>[127.0, 150.0]</b> |
|                      | Median difference<br>[95% CI] |                          | <b>-0.7</b><br><b>[-4.7, 2.8]</b>     | 21.6<br>[19.0, 26.7]             | <b>-3.3</b><br><b>[-5.0, -0.9]</b>    |



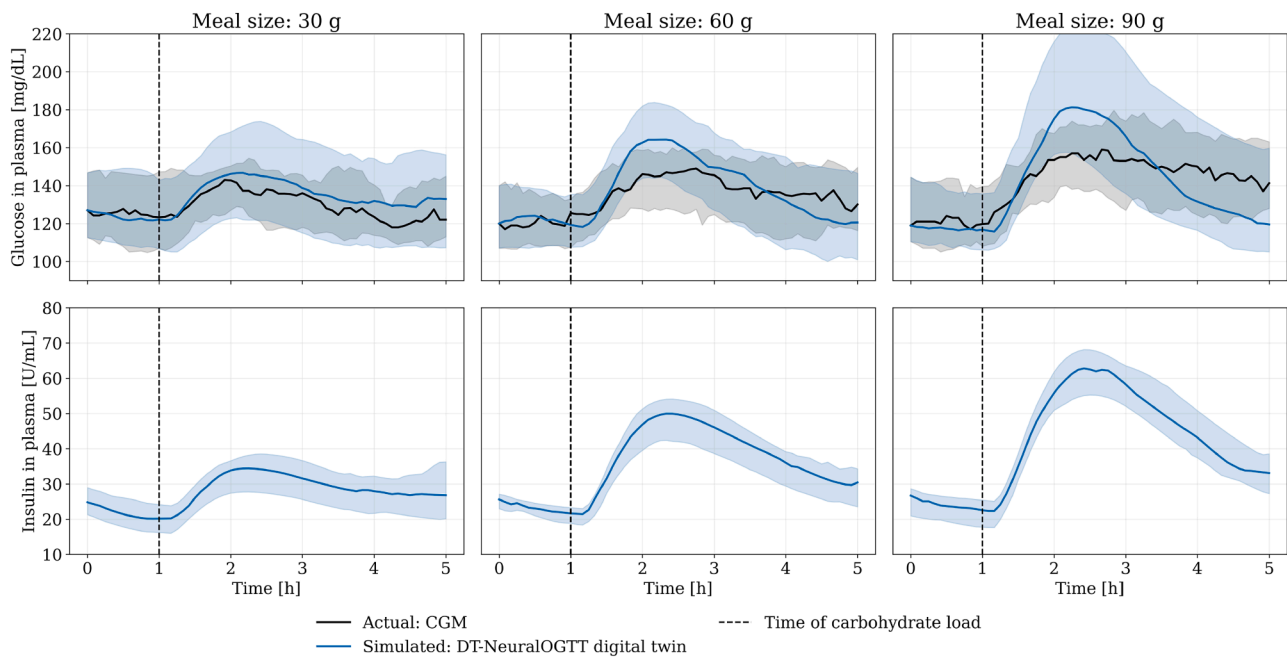
**Fig. 6.** Box plot for the distribution of the average root mean squared error (RMSE) across varying length simulation of the DT-NeuralOGTT digital twins, from 30 min up to 12 h utilizing only one initialization point at  $t = 0$ . Horizontal lines reflect the median, while circles reflect the mean.

### 3.2.1. Impact of individual-level inputs and model response to controlled perturbations

Supplementary Material S9 presents the results of the ablation study, showing model performance across meal, physical activity, and sleep scenarios when trained with different subsets of individual-level inputs. All model configurations met the equivalence criteria across all evaluated glucose metrics. This indicates that the model is capable of

adapting to CGM data patterns even in the absence of external contextual inputs beyond carbohydrate intake and insulin (when available). However, this does not imply that individual-level inputs are unnecessary.

The value of incorporating the full set of inputs becomes evident when evaluating the model under controlled input perturbations. In these scenarios, the DT-NeuralOGTT digital twins demonstrate consis-



**Fig. 7.** Simulated glucose and insulin responses under standardized meal perturbations for DT-NeuralOGTT digital twins. The top row shows glucose responses and the bottom row shows insulin responses following carbohydrate loads of 30 g (left), 60 g (center), and 90 g (right). The dashed vertical line indicates the time of carbohydrate ingestion. Median simulated trajectories are shown as solid blue lines, with shaded regions representing the interquartile range (25th–75th percentile) across individuals. Actual CGM data are shown in black for reference in the glucose panels. Simulations are displayed over a 4-h window to capture typical postprandial dynamics.

tent and physiologically plausible responses to changes in key drivers of glucose dynamics.

For meal-related scenarios, Fig. 7 shows the simulated responses to standardized carbohydrate intakes of 30 g, 60 g, and 90 g. The top row presents the glucose responses, while the bottom row shows the corresponding insulin dynamics. From left to right, each column represents increasing meal size, with real CGM data shown in black as a reference. As the meal size increases, the glucose response exhibits a proportional increase in both peak magnitude and duration. Correspondingly, insulin response also increases, reflecting appropriate physiological regulation. Although direct insulin measurements are not available for comparison, the simulated insulin trajectories remain within clinically plausible ranges.

To evaluate the contribution of heart rate-based features, we analyzed self-reported physical activity events from the testing set. Fig. 8 compares simulations with actual heart rate features and with heart rate replaced by baseline values. Differences on the glucose response are observed across all activity durations, highlighting the importance of heart rate information in capturing glucose dynamics.

When heart rate features are included, the model reproduces the expected physiological response, with glucose reductions during or after physical activity and corresponding modulation of insulin dynamics. In contrast, removing these features leads to attenuated and less temporally aligned responses. These differences are more pronounced for longer activities (> 40 min), where the full model more closely matches observed glucose patterns, indicating that heart rate features are particularly important for capturing sustained effects of physical activity. Notably, for short-duration activities (10–40 min), the model tends to underestimate the rapid glucose decrease observed in the CGM data; however, it still provides a closer approximation than the model without heart rate features.

To evaluate the effect of medication inputs, we analyzed 12-h simulations under controlled perturbations for exogenous insulin, sulfonylureas, and biguanides, as shown in Fig. 9. From left to right, the figure presents results for each medication class. For exogenous insulin, the effect on glucose dynamics is clearly captured, with an increase in sim-

ulated plasma insulin followed by a corresponding reduction in glucose levels. In contrast, sulfonylurea and biguanide effects are more modest but still observable. In particular, sulfonylureas exhibit a delayed response, with noticeable glucose reduction occurring approximately 6–8 h after administration.

While the model captures appropriate glucose trends, the effects of GLP-1 and SGLT-2 medications were not accurately reproduced, as no clear glucose response was observed under perturbation. Nonetheless, these results demonstrate the flexibility of the proposed hybrid framework, which can incorporate multiple medication inputs within a physiologically structured model.

### 3.2.2. Multi-day simulations

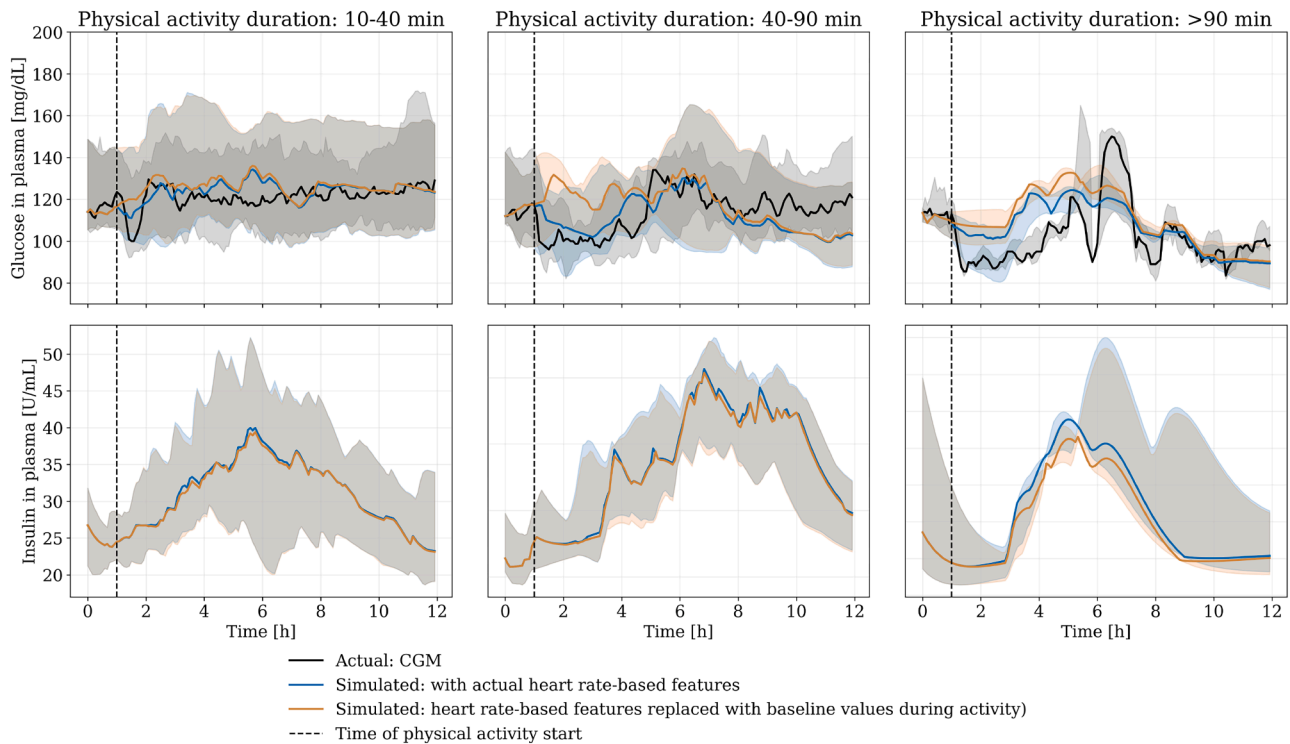
We illustrate the models' simulation capability with a 3-day example in which only the initial CGM values and the CGM trend of the last 30 min were used for initialization (Fig. 10). Additional results for the virtual patient population are provided in Supplementary Material S10, including a 3-day example for each digital twin and RMSE and glucose outcomes for 12-h test sequences. These findings show that, although the models were trained on 12-h sequences, the simulation window can be extended, though further work is needed to define its limits.

## 4. Discussion

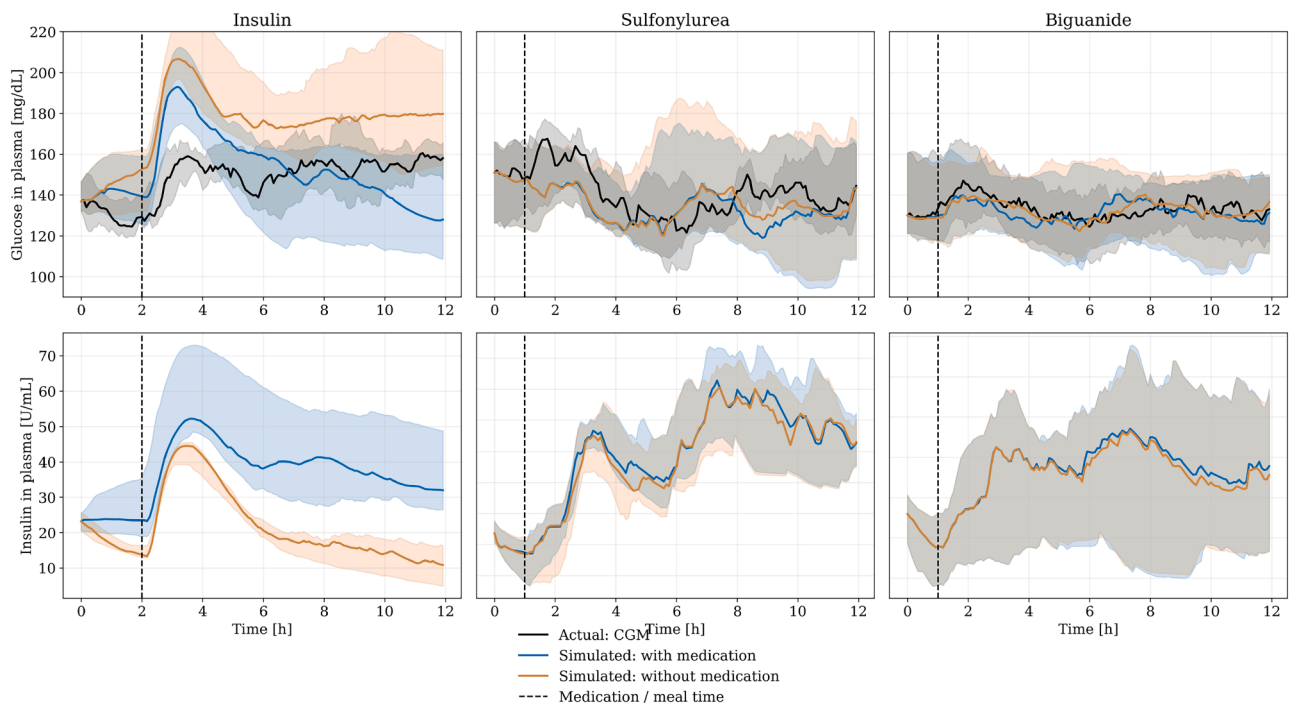
### 4.1. Overview of the proposed digital twin framework

We developed a framework for creating digital twins that simulate glucose dynamics in pre-T2D and T2D. Our approach combines a glucose-insulin model in the context of the OGTT (NeuralOGTT) with individual-level models to construct digital twins that capture residual dynamics not modeled by NeuralOGTT (DT-NeuralOGTT). The individual-level models can be trained to represent new inputs such as time of day, physical activity, stress, and medications typically used for treating diabetes, but which are not modeled by ODEs.

We used variance-based sensitivity analysis to simplify the structure of NeuralOGTT, while preserving model performance and explainabil-

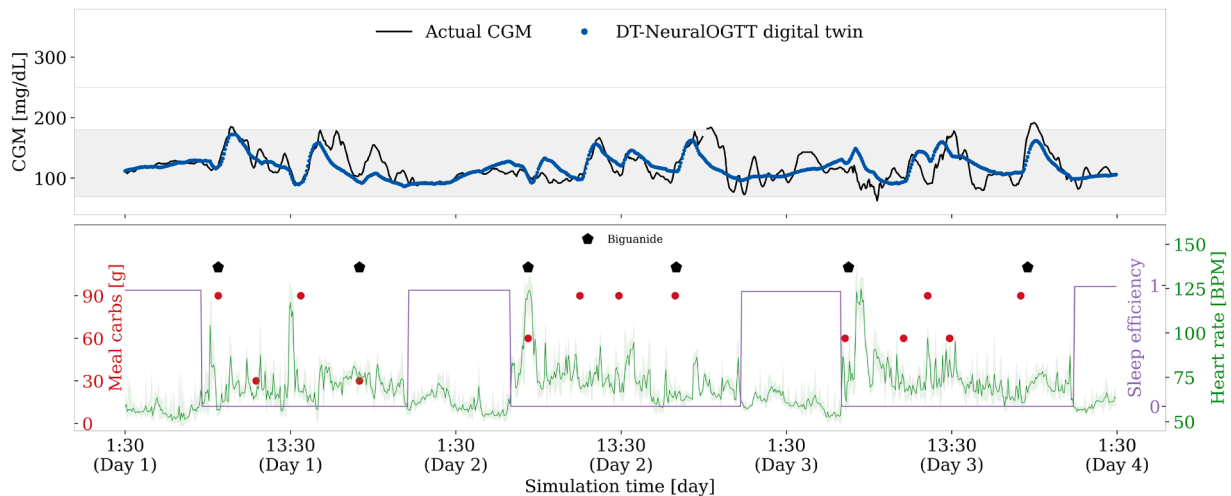


**Fig. 8.** Effect of heart rate-based features on simulated glucose and insulin responses during physical activity. The top row shows glucose responses and the bottom row shows insulin responses for physical activity events grouped by duration: 10–40 min (left), 40–90 min (center), and >90 min (right). The dashed vertical line indicates the start of the physical activity. The black line represents the actual CGM data. The blue line corresponds to DT-NeuralOGTT simulations using the actual heart rate-based features, while the orange line shows simulations where heart rate features were replaced with baseline values. Solid lines represent the median response and shaded regions indicate the interquartile range (25th–75th percentile).



**Fig. 9.** Effect of medications on simulated glucose and insulin responses. The top row shows glucose responses and the bottom row shows insulin responses for: exogenous insulin (left), sulfonylurea (center), and biguanide (right). The dashed vertical line indicates the moment where the medication was taken. The black line represents the actual CGM data. The blue line corresponds to DT-NeuralOGTT simulations using the actual medication dose, while the orange line shows simulations where medication was removed. Solid lines represent the median response and shaded regions indicate the interquartile range (25th–75th percentile).





**Fig. 10.** Example of a 3-day glucose trace from the actual CGM vs. AI-based digital twin (Top panel). To demonstrate utility, we choose a scenario that is challenging to replicate as our example, which includes doses of biguanide, carbohydrate intake, heart rate-base features, and sleep efficiency (Bottom panel). The simulation starts at 1:30 am, with a single initialization at the beginning. *BPM: Beats per minute.*

ity. We found that NeuralOGTT consistently performs well even on unseen glucose loads despite being trained on a small dataset, highlighting the utility of a hybrid modeling approach whereby domain knowledge in the form of a physical model can be used to guide the learning phase of a data-driven model. This observation demonstrates the physiology-based constraints imposed during training, which allowed the model to learn causal relationships and enable its further expansion to a virtual patient population which may ultimately be used to design and test new decision support and automated hormone delivery interventions.

Although DT-NeuralOGTT is implemented using NNs, the overall model remains interpretable because each state corresponds to a well-defined physiological process. This ensures that the model's observable input–output behavior is consistent with glucose–insulin physiology. Therefore, it can be used for *in-silico* testing, for example, to assess the effectiveness of recommendations generated by a decision support system for meals or physical activity, since different scenarios can be simulated and the resulting glucose responses evaluated for each individual. This is further highlighted by the sensitivity analysis provided herein.

#### 4.2. Comparison with other modeling approaches

These results are further supported by the comparison with alternative modeling approaches. As expected, the data-driven model showed strong agreement with the observed data, reflecting its ability to directly learn patterns from the data. However, such models lack physiological interpretability and may fail to generalize under changing conditions. Conversely, the mechanistic model [20], while physiologically interpretable, proved more difficult to individualize due to the complexity of tuning parameters that must simultaneously capture multiple interacting physiological processes.

In this context, DT-NeuralOGTT provides a balanced approach, combining the personalization capacity of data-driven methods with the interpretability of mechanistic models. This advantage becomes particularly evident when analyzing the role of individual-level inputs. While the model can reproduce the glucose outcomes for the *T2Help Data – Free Living* even with reduced input configurations, controlled perturbation experiments show that inputs such as heart rate and medications are critical to capture realistic physiological responses under varying conditions. In particular, these inputs enable the model to appropriately respond to external stimuli (e.g., meals, physical activity, and medications) improving its ability to capture underlying dynamics rather than average behavior. Together, these findings highlight the benefit of the proposed hybrid framework, which leverages both physiological struc-

ture and data-driven learning to improve simulation fidelity and robustness.

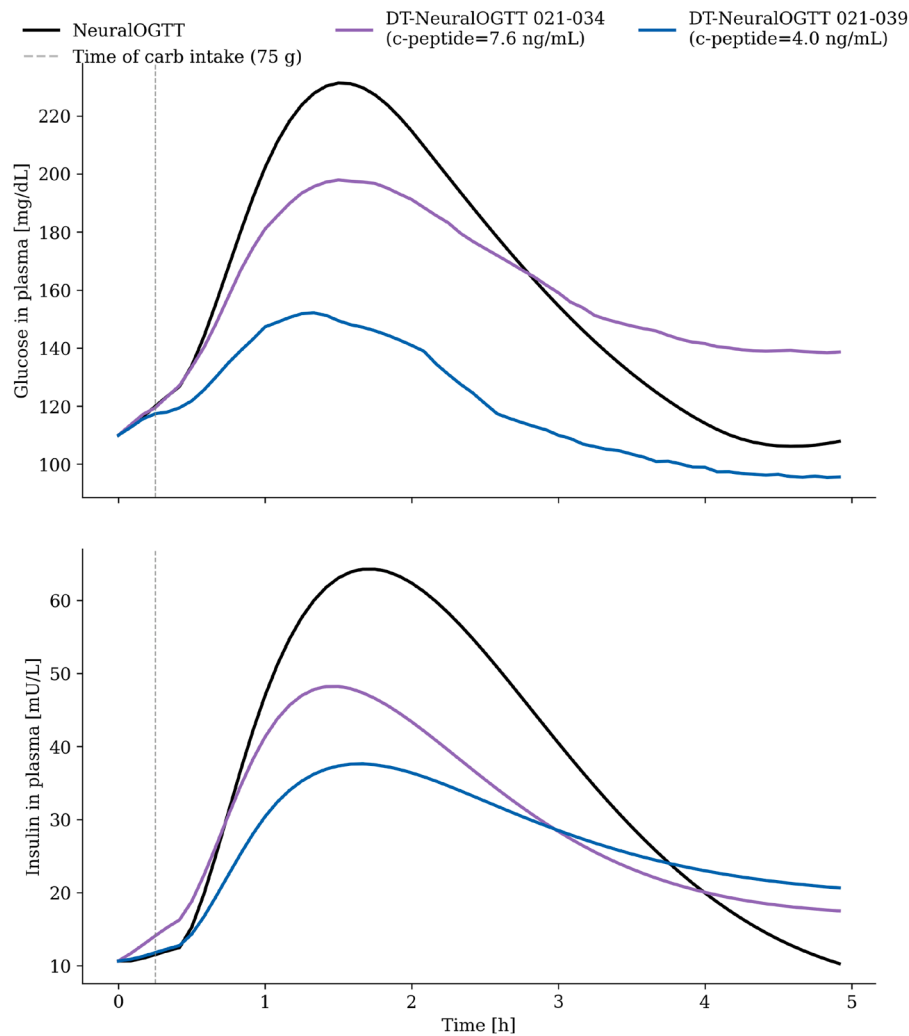
#### 4.3. Accuracy of the virtual patient population

A strength of our model lies in the definition of the loss function and the search for the optimal individual penalization parameters for hypoglycemia and extreme hyperglycemia events, allowing the resulting virtual patient population to be a good representation of metabolism in people with pre-T2D and T2D. This is evidenced by the Parkes error grid, which indicates that 98.8% of the simulated glucose values fall within the A + B region of the testing set, underscoring the model's accuracy and suitability as a potential clinical tool. Notably, even for extended 12-h simulations, considerably longer than the achieved with other published models [15–19], the simulator demonstrates very high short-term accuracy and accuracy that plateaus across the 12-h simulation window (Fig. 6) while preserving glucose outcome distributions that remain highly similar to the actual data (Table 4).

Each digital twin from the virtual patient population consistently performs well despite the heterogeneity among individuals living with pre-T2D and T2D, considering factors such as medication therapy and lifestyle. This performance is achieved because our approach relies on personalization. If our approach relied solely on combining NeuralOGTT with a model trained on all T2Help data, it would fail to capture inter- and intra-individual variability. The dataset is small ( $N = 38$ ) and includes participants at different disease stages, from pre-T2D to insulin-dependent T2D. As a result, such a model would learn an averaged response rather than realistic individual glucose dynamics, which is why, in this case, an individualized modeling approach is more appropriate. It is also remarkable that, given the flexible definition of the model, the digital twin can continuously evolve alongside the real-world twin as the disease progresses and as more data becomes available.

#### 4.4. Interpretation of the role of the individual-level model in DT-NeuralOGTT

The role of  $N^{\text{Ind}}$  is, by definition, to compensate for aspects of glucose regulation that are not fully captured by the frozen NeuralOGTT component. During personalization, the individual-level model learns adjustments to the state derivatives that improve agreement with the observed CGM trajectories. Since only glucose measurements are available during this phase, the model cannot uniquely determine which physiological mechanism is responsible for a given correction. Consequently,



**Fig. 11.** Comparison of NeuralOGTT and personalized digital twins in participants representing different metabolic states. Simulations were initialized from identical conditions and included a 75-g carbohydrate intake administered 15 min after the start of the simulation. Results are shown for the frozen NeuralOGTT model (black), a participant with the highest C-peptide value in the cohort (7.6 ng/mL, purple), and a participant with a C-peptide value of 4.0 ng/mL who was classified as pre-diabetic (blue). The top panel shows plasma glucose, and the bottom panel shows plasma insulin.

we cannot definitively separate whether a correction is compensating for a limitation of the baseline NeuralOGTT representation or capturing genuine inter-individual physiology. We recognize this as an inherent limitation of the current framework.

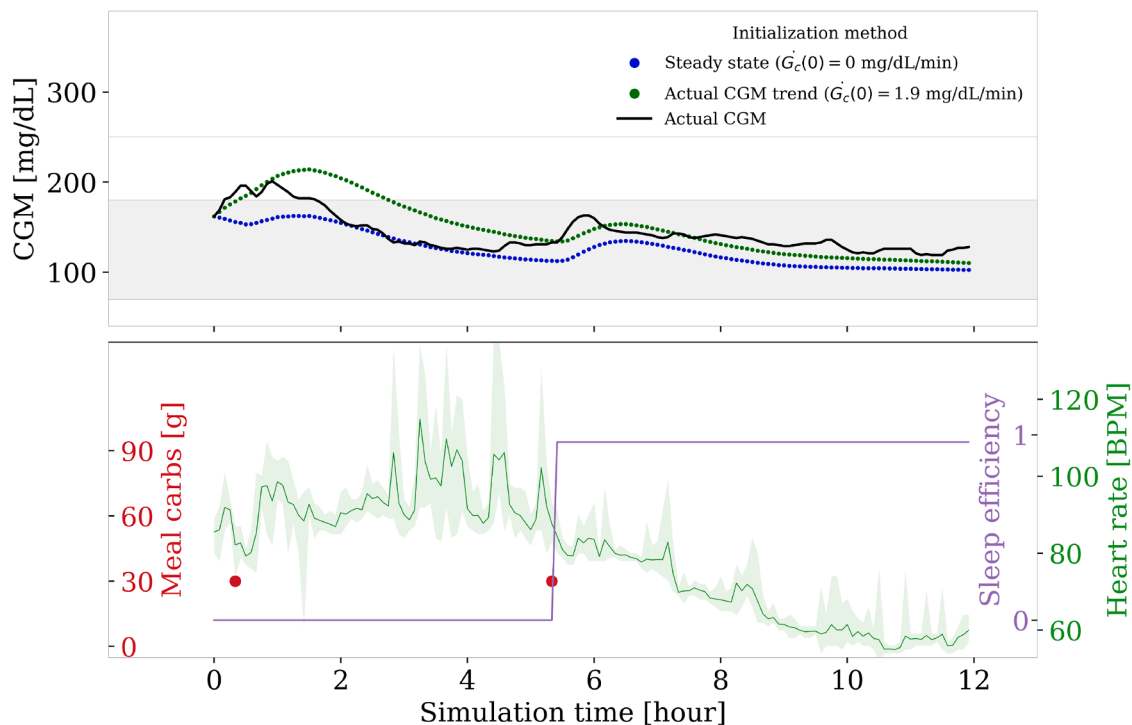
To better understand this behavior, we selected two digital twins representing different metabolic states within the study cohort. The first corresponds to the participant with the highest observed C-peptide level (7.6 ng/mL), which may reflect greater insulin resistance and compensatory endogenous insulin production. The second corresponds to a participant classified as pre-diabetic with a C-peptide level of 4.0 ng/mL. Neither participant was receiving exogenous insulin therapy. By comparing these individuals, we sought to evaluate how the personalization component  $N^{\text{Ind}}$  acts on the frozen NeuralOGTT baseline and adapts it to represent participant-specific glucose-insulin dynamics.

Specifically, we simulated a 75-g carbohydrate intake using the frozen NeuralOGTT and the corresponding personalized digital twins. Fig. 11 presents the results for both digital twins. Relative to NeuralOGTT, both individuals exhibit lower glucose and insulin responses, which is expected because the digital twins were trained from free-living carbohydrate intake data rather than OGTT glucose intake experiments. More importantly, the participant with the highest C-peptide level exhibits higher postprandial glucose and insulin responses than the participant with lower C-peptide levels, consistent with the greater

insulin resistance typically associated with elevated endogenous insulin production. In both cases, the individualized model modifies both glucose and insulin dynamics while preserving physiologically plausible behavior.

We would also like to emphasize an additional challenge when personalizing the framework for individuals receiving exogenous insulin therapy. During training, the personalized model only observes CGM glucose measurements together with the recorded exogenous insulin administration. Because no data are available describing how the individual would respond to carbohydrate intake in the absence of exogenous insulin, the model cannot directly disentangle endogenous insulin secretion from the effects of administered insulin. As a result, the corrections learned by  $N^{\text{Ind}}$  within the insulin compartment may not correspond to the individual's true physiological insulin response. Instead, part of the learned correction may compensate for this identifiability limitation, making the insulin-related latent states less interpretable than in individuals who are not receiving insulin therapy.

Consequently, while the framework can account for exogenous insulin inputs and reproduce observed glucose dynamics, the physiological interpretation of the personalized insulin trajectories should be treated with caution in these individuals. This represents an important limitation of the current approach. Future work could improve both interpretability and physiological fidelity through the incorporation of ad-



**Fig. 12.** Example of the effect of initialization method in a 12-h simulation for two approaches. Green points indicate simulations using the actual CGM trend over the last 30 min, while blue points represent simulations using steady-state assumptions, where the CGM trend is assumed to be zero. Initializing with the actual CGM trend at the time of initialization allows the model to capture peaks that were not detected when the trend was set to zero.

ditional measurements related to endogenous insulin secretion, such as C-peptide. Such information would provide direct supervision of the insulin compartment, helping distinguish endogenous insulin production from exogenous insulin effects and enabling more reliable personalization in individuals receiving insulin therapy.

#### 4.5. Future work and limitations

While this work provides many novel points of interest, we emphasize some areas where future work may lead.

First, the accuracy of the simulations is highly dependent on the initialization of the model states. Our decision to incorporate the CGM trend from the 30 min prior to initialization as a constraint may have contributed significantly to the success of the methodology. Fig. 12 illustrates how initialization of the states impacts simulations. Specifically, when states are initialized using the CGM trend from 30 min prior, the simulation successfully captures a glucose level above 180 mg/dL at the beginning of the 12-h window. In this example, the effect of the initialization method continues to influence glucose levels even 5 h into the simulation, where a second meal entry occurs, the simulation that uses the actual CGM trend closely matches the real data, whereas the one using steady-state assumptions does not.

However, this strong dependence on recent CGM data introduces a practical limitation. The digital twin requires a continuous, uninterrupted 30 min of CGM signal prior to initialization to ensure reliability, which may restrict its applicability in real-world scenarios where data gaps, sensor dropouts, or missing measurements are common. As suggested by Fig. 12, the impact of initialization errors can persist for several hours, further amplifying this constraint.

We acknowledge that state initialization remains a challenging aspect of this work and an area with significant room for improvement. Future efforts should focus on developing more robust initialization strategies that can handle incomplete or noisy CGM data, such as data imputation techniques, probabilistic state estimation, or learning-based initialization methods [46]. Enhancing this component would improve not

only simulation outcomes but also the robustness and practical usability of the overall framework.

Second, the T2Help cohort includes predominantly well-controlled individuals with T2D, as reflected by the relatively high median TIR of approximately 92%, which exceeds typical clinical targets. This indicates that the cohort may not be fully representative of the broader T2D population, particularly individuals with below target glycemic control, potentially limiting the generalizability of the proposed digital twin framework. The need for validation across more diverse populations has been identified as a critical challenge for the broader adoption and clinical translation of digital twin technologies [47]. Nevertheless, the dataset still includes participants with less optimal glycemic control, such as participant 021-015, for whom the model demonstrates good performance. Detailed performance metrics at the individual level are provided in Supplementary Material S10, supporting the robustness of the approach across a range of glycemic profiles. While future studies should aim to validate the framework in more diverse and representative populations, the present results demonstrate the potential of the proposed methodology as a valuable tool for personalized glucose modeling.

Third, this model relies on data completeness for heart rate and sleep efficiency, excluding sequences that do not meet certain criteria. In future work, we may consider mechanisms for data imputation in cases where this data is incomplete [48].

Fourth, our evaluation is based on CGM-derived control metrics, which are standard and clinically meaningful measures of glycemic control in pre-T2D and T2D. Additional physiological indicators, such as insulin sensitivity or resistance, could further strengthen the assessment of physiological fidelity. Although the model includes compartments representing insulin production and insulin dynamics, we do not have the corresponding physiological measurements needed to validate the accuracy of this estimated physiological marker.

Lastly, one limitation of our individual-level model is our reliance on self-reported meals and medication events. While individuals strive to accurately report their data, it is not guaranteed to reflect ground truth.

In future iterations we may explore alternatives such as inclusion of a meal detection algorithm.

## 5. Conclusions

This work introduces NeuralOGTT and DT-NeuralOGTT, the first physiology-guided AI and digital twin models capable of forecasting glucose outcomes in pre-T2D and T2D while capturing inter- and intra-individual variability with minimal data. This hybrid approach, combining physiological models with NNs, improves *in-silico* studies by enabling more accurate and personalized analyses using small datasets.

We also contribute methodological advances for training AI models. First, our variance-based sensitivity analysis offers a new way to select model architecture that balances accuracy with physiological interpretability. Second, we develop a robust, customizable loss function that supports person-centered modeling.

Together, these innovations establish a framework for more personalized, data-efficient, and interpretable AI approaches in diabetes research and clinical glucose management.

## CRediT authorship contribution statement

**Valentina Roquemen-Echeverri:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Clara Escorihuela-Altaba:** Writing – review & editing, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization; **Taisa Kushner:** Writing – review & editing, Conceptualization; **Peter G. Jacobs:** Writing – review & editing, Conceptualization; **Jose Garcia-Tirado:** Writing – review & editing, Methodology, Conceptualization; **Clara Mosquera-Lopez:** Writing – review & editing, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

## Declaration of competing interest

Peter Jacobs reports advisory board participation and research support from Eli Lilly and Dexcom. Peter Jacobs have a financial interest in Pacific Diabetes Technologies, a company that might have a commercial interest in the results of this research and technology. All other authors declare no conflicts of interest.

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## Supplementary material

Supplementary material associated with this article can be found in the online version at [10.1016/j.bbe.2026.08.001](https://doi.org/10.1016/j.bbe.2026.08.001).

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